

Factors impeding growth of the industry

- Small and medium enterprises (SMEs) and startups lack experience in complex system design. They are completely unaware of outsourcing opportunities (both design and manufacturing).
- Computer-aided design (CAD) tools are expensive, and open source tools are not of acceptable quality. Engineering design is capital-intensive. Educational institutes that have relevant tools (CAD as well as T&M equipment) could work out a scheme to offer labs-on-hire. This would help startups and SMEs to develop products at a low cost.
- There is a dearth of hardware accelerators supporting board design and FPGAs/EPLDs.
- There are governance/policy-related challenges, such as:
 - Prototype ecosystem is yet to mature, leading to a lot of wasted time in customs and other processes.
 - Duty-free components are still being charged at 28 per cent, which should be reduced.
 - While there is a steady supply of graduates in the market, there is still a significant gap between skill sets required by the industry and skill sets acquired by graduates through higher education institutes. Effective collaboration programmes between industry and institutes can go a long way in developing more employable candidates.

margins are already thin. Pressed for margins, firms prefer to shop for products that are suitable for the Indian market and arrange for imports of the semi-knocked-down kits back in India for assembly. This has had a huge impact on the design industry.

Policy boost

The government of India is taking a strong look at these issues, and is trying to develop policies and frameworks that can catalyse growth in the design sector. Developing policies such as Make in India and Digital India, and providing impetus to existing schemes like modified special incentive package scheme (M-SIPS) and electronic development fund (EDF) will likely have significant impact in driving change.

The industry has responded positively to the government's support. One hundred and fifty nine new ESDM units were established in 2015 itself. The industry also feels, while Make in India is a vision in the right direction, the agenda could also include Make for India with a focus on Indian priorities.

Other supportive measures taken by the government include:

- Hundred per cent foreign direct investment (FDI) allowed in the electronics hardware manufacturing sector under automatic route
- Duty relaxation and schemes—such as EPCG, EHTP and SEZs—to

provide tax sops; duty exemption for equipment required for setting up semiconductor plants

- National Policy on Electronics (2012) and setting up of National Electronics Mission

Moving forward

New technologies are getting adopted faster worldwide, including India. However, the Indian market is unique with respect to application areas as well as cost scenario. This, in turn, develops demand for new product development to cater to specific needs of Indian markets across application areas, including telecom, automotive, medical and industrial products.

Key trends that will transform the electronics design space for new-age applications are mobility, cloud and analytics. Emerging application areas include industrial and automotive microcontrollers, wireless infrastructure segments, consumer digital TVs, handheld devices and compute/storage infrastructure.

Moreover, special design services will be required for smart energy management, medical telemetry applications and high-end application processor development.

All these emerging trends in electronics design services indicate collaboration between manufacturing and design industries, so that Design in India can lead to Make in India. **EFY**

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